

Quantifying Reef Fish Color Patterns

An Introduction to pavo • Lab 5 • Week 6

The big question

Given a fish photo, can we put numbers on its pattern?

We can describe color patterns in words — "barred," "spotted," "reticulate." But to compare species, test hypotheses, or build phylogenetic models, we need numbers we can compute and replicate.

That is what pavo's adjacency framework gives us.

Today's three demo fish — three patterns, three families



Sergeant major

Pomacentridae

vertical bars



Cleaner wrasse

Labridae

horizontal stripe



Mandarin fish

Callionymidae

reticulate maze

Three fish that look different — let's see if pavo can SEE that difference in numbers.

Step 0 — Segment the fish from its background

Raw photo



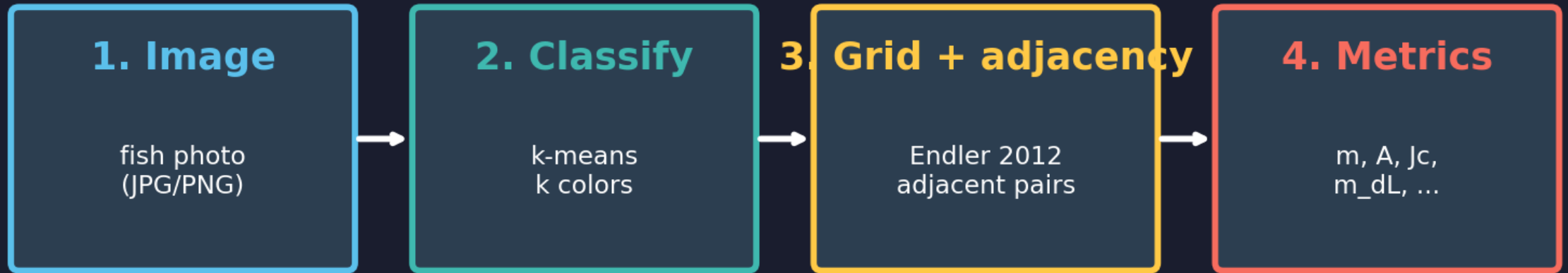
Segmented (background removed)



`segment_fish()` uses `rembg` to isolate the animal so pavo never classifies background pixels as a color class.

Without this step, k-means would find "reef-blue" or "sand" as one of your fish's color classes — and your pattern statistics would describe the photo, not the animal.

The pavo pipeline — four steps



We're going to walk through each box in detail. The whole point is to demystify what's happening at each step.

Step 1 — Load the image

`getimg()` reads JPG / PNG into a 3D array

What does pavo see when it loads an image?

An image is a 3D array of numbers: width $\times$ height $\times$ {R, G, B}.

A 1280×720 photo holds $1280 \times 720 \times 3 \approx 2.8$ million numbers.

R code

```
library(pavo)

fish <- getimg("cleaner-  
wrasse.jpg")  
plot(fish)
```

What you see



Step 2 — Classify the colors

k-means reduces millions of unique RGB values to k color classes

Why classify? — The 16-million color problem

- An RGB image has up to $256^3 = 16.7$ million possible colors per pixel.
- Within one fish photo, there might be 50,000 unique RGB values.
- But biologically, there are only a handful of distinct color regions — maybe 3 (yellow, black, white) or 5 (multi-band angelfish).
- We need to collapse the continuous color cloud to a small, discrete palette.
- That palette is the input to all the pattern statistics.

k-means in one paragraph

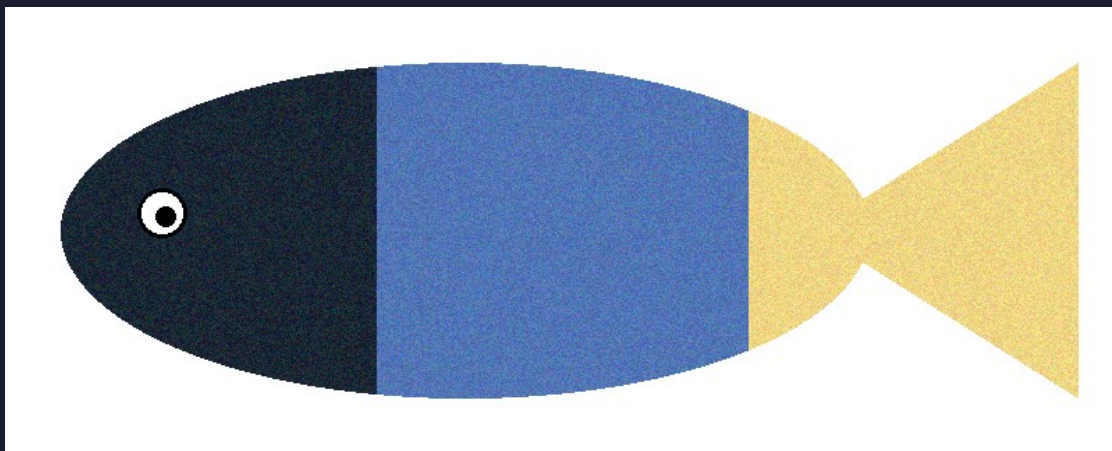
Pick k . Drop k random points into RGB space. For each pixel: snap to the nearest of the k points. Then move each of the k points to the centroid of the pixels assigned to it. Repeat until nothing moves. The k final points are your color classes.

That's what `classify(img, kcols = k)` does inside pavo.

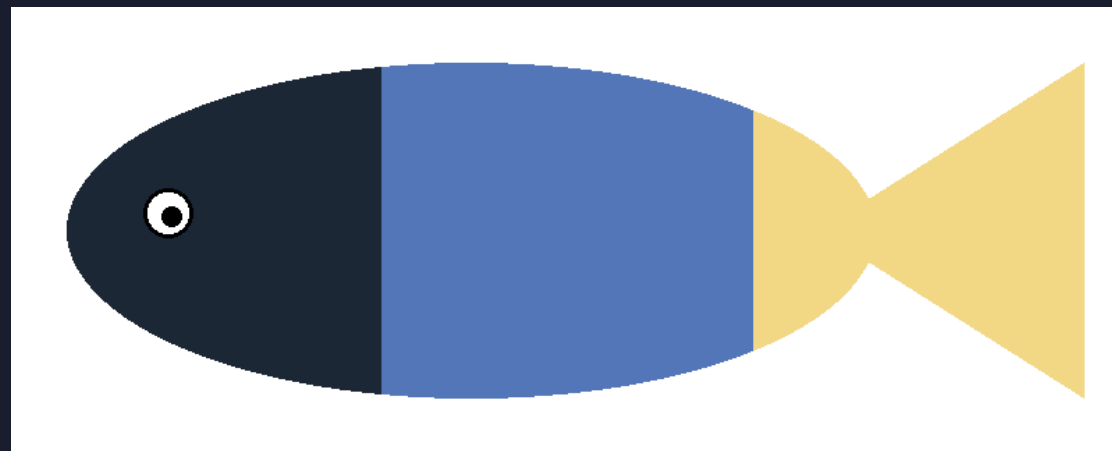
```
fish_k3 <- classify(fish, kcols = 3)
plot(fish_k3)
```

classify() — before and after, $k = 3$

Before — natural variation



After — classified to $k = 3$



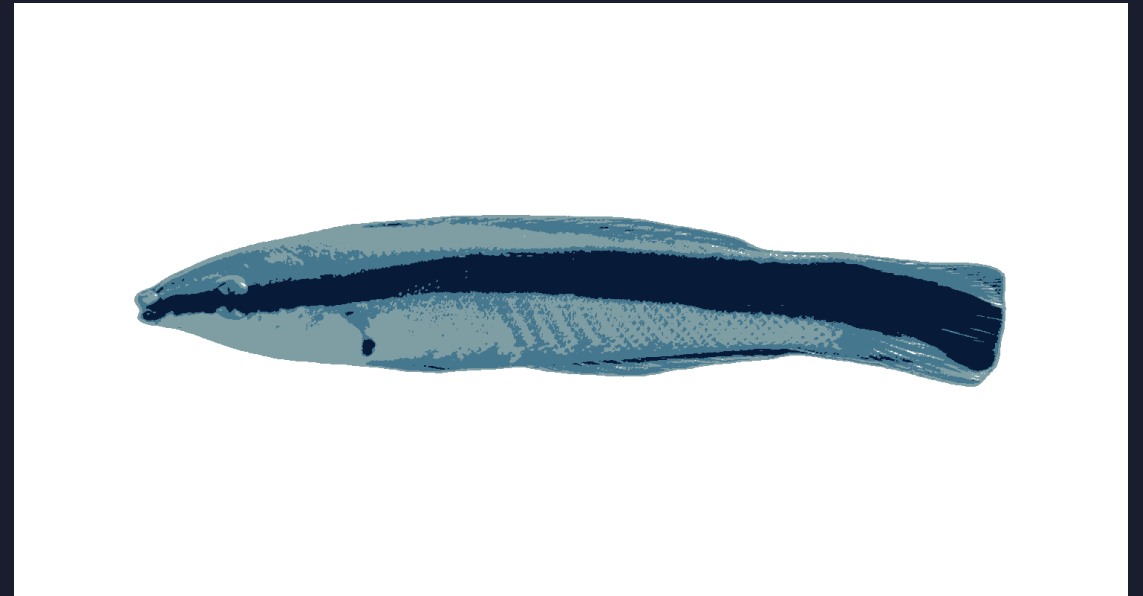
Every pixel is now ONE of 3 color labels. The continuous color cloud has become a categorical map.

classify() on a real fish — cleaner wrasse, $k = 4$

Before — full color photograph



After — classified to $k = 4$



Same idea on a real fish. Every pixel collapses to one of 4 color labels — black mask, yellow back, white belly, white background.

How do you choose k ?

- Eyeball it: how many distinct color regions does the fish have?
- Try $k = 3, 4, 5$; plot each; pick the one that looks like the biology.
- There's no single right answer — pavo respects what the analyst decides.
- For a 2-color animal (yellow tang, all-black drum), $k = 2$ is fine.
- For multi-band angelfish or pufferfish, you may need $k = 6$.
- Today: sergeant major $k = 3$, cleaner wrasse $k = 3$, mandarin $k = 5$.

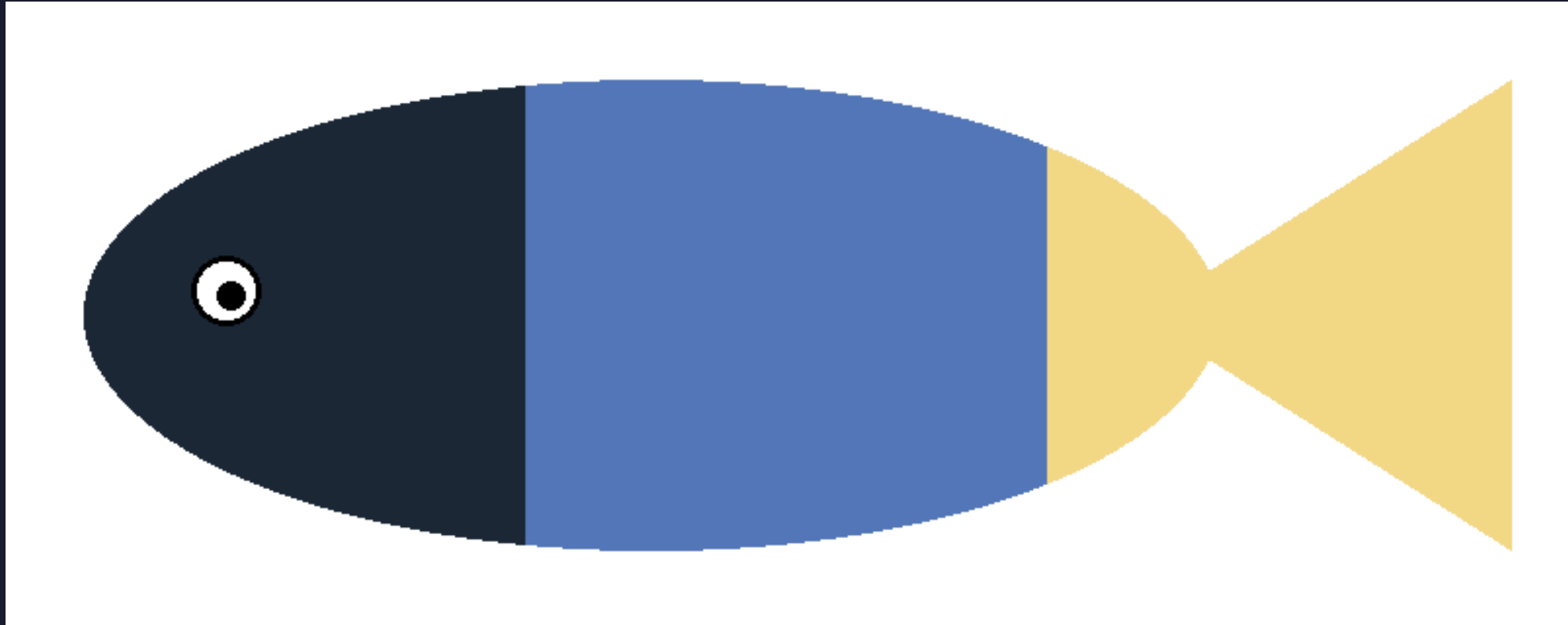
Step 3 — Lay down a grid

The Endler 2012 idea: sample the pattern at regular points, ask what colors are adjacent at each point.

Why a grid?

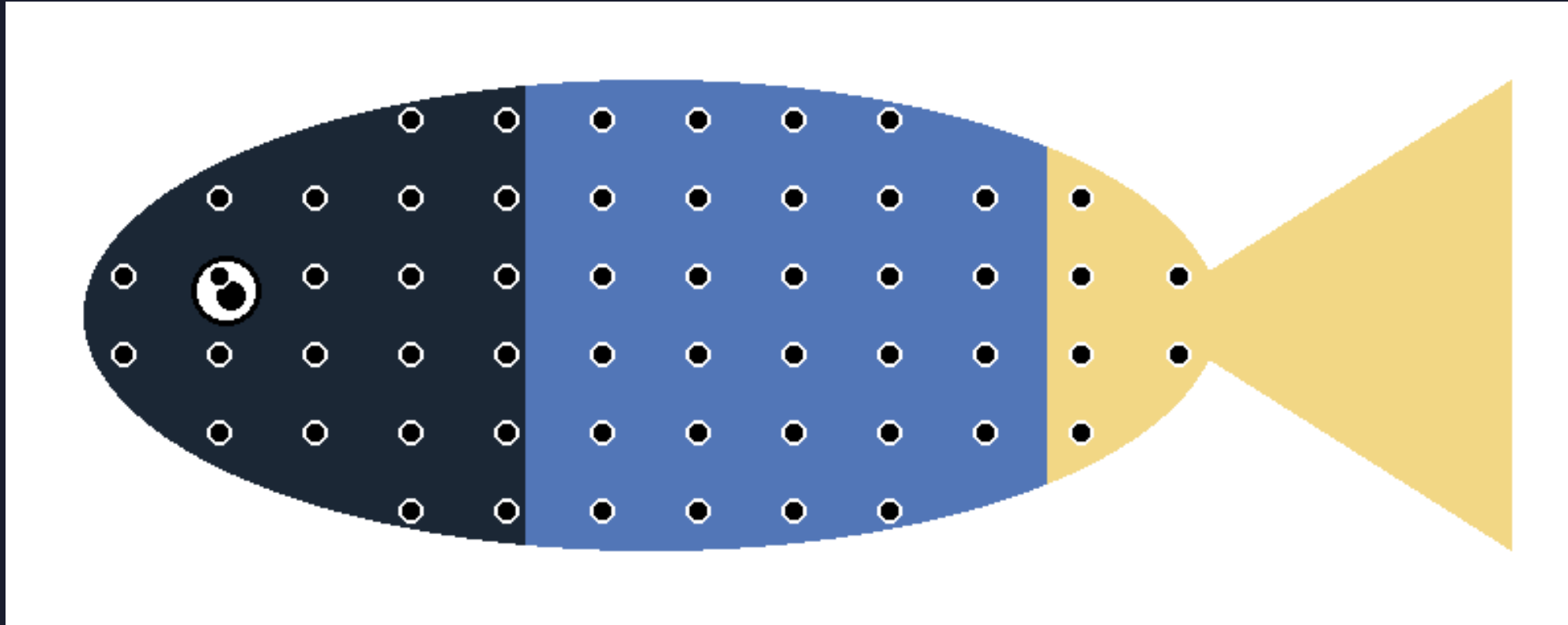
- We want to summarize the pattern statistically — not pixel-by-pixel.
- Endler (2012) proposed a regular grid of sampling points across the patch.
- At each grid point, compare the focal pixel to its cardinal neighbors.
- Tally how often color A is next to color B → a transition matrix.
- Every metric pavo reports is a function of that transition matrix.
- Same algorithm works on a fish, a frog, a flower, a feather. Generalizable.

Frame 0 — Start with the classified fish



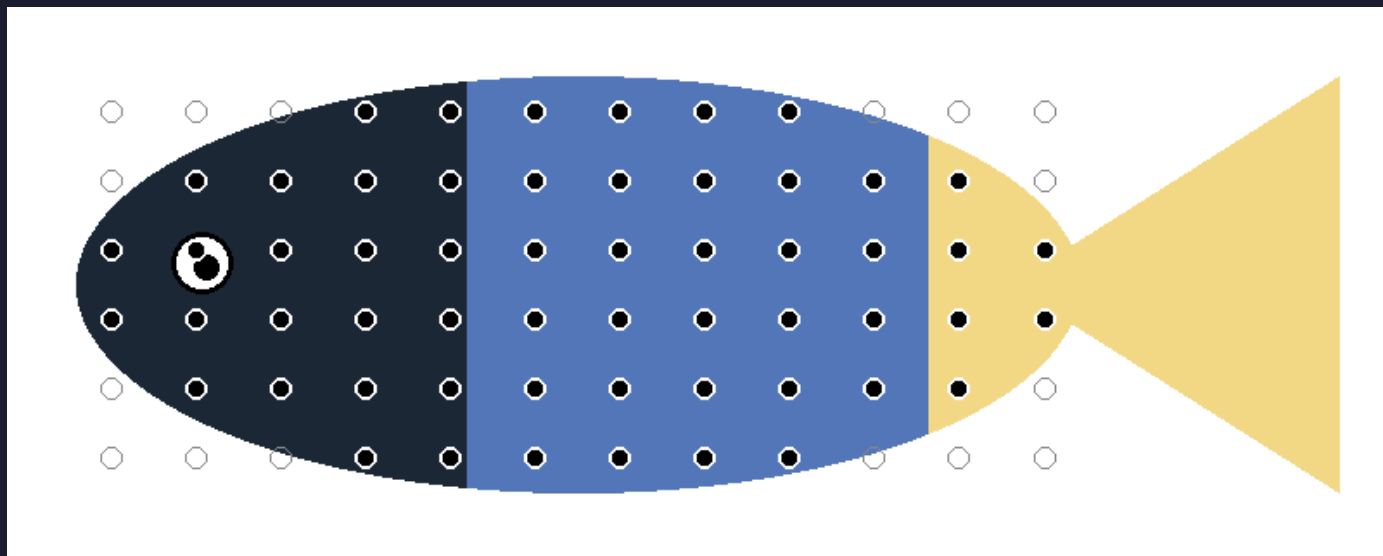
Three color classes: dark, mid, light. No grid yet.

Frame 1 — Drop a regular grid of points



Grid is laid down on the FISH only. Background points are excluded (see next slide).

How pavo handles the background — and edge points



Hollow grey rings = grid points pavo DROPS
(they fall on the white background)

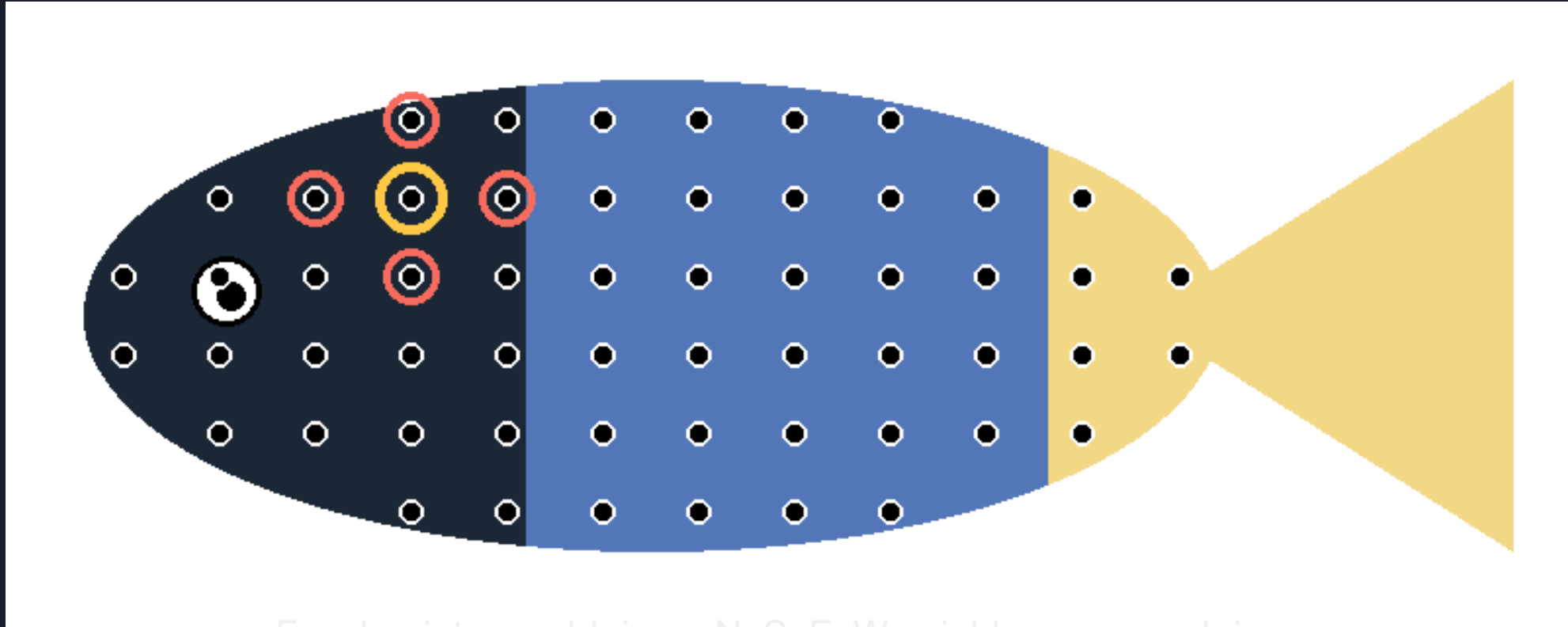
bkgID — flag the background class

After `classify()`, one of the k color classes is the white padding around the fish. Pass that class number to `adjacent(bkgID = X)`. pavo then drops every transition that touches the bkg class — both the focal pixel and any of its neighbors that fall on background.

Points with < 4 neighbors

Edge / contour points may only have 2 or 3 valid neighbors (the others fall off the body or on bkg). pavo simply tallies whatever transitions ARE valid — the matrix is then normalized to proportions, so a point with 2 neighbors contributes $\frac{1}{2}$ as much weight as one with 4. No extrapolation, no padding, no fudge.

Frame 2 — Pick ONE point. Look at its 4 cardinal neighbors.

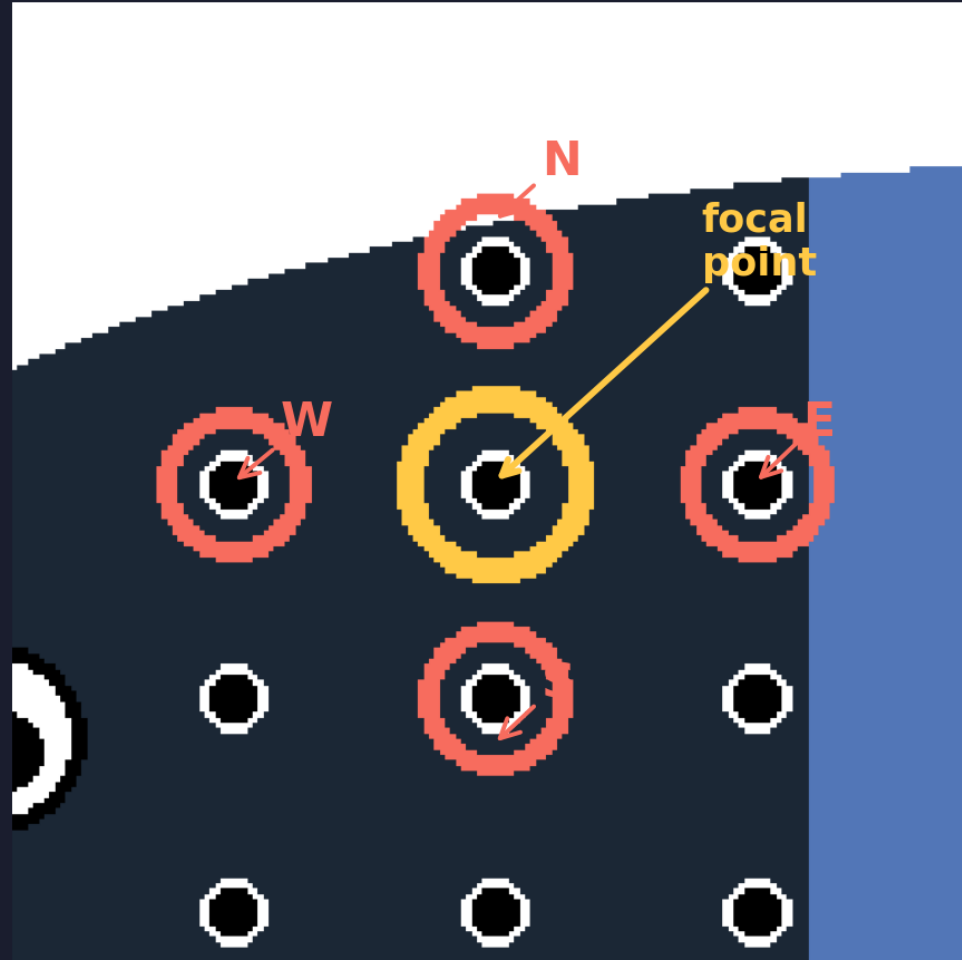


Focal point — gold ring. N, S, E, W neighbors — coral rings.

We are going to count: what color is the focal pixel? What color is each neighbor?

Frame 3 — Zoom in on the focal point

At each grid point: examine focal pixel + its 4 cardinal neighbors



One focal pixel + 4 neighbors = 4 adjacency tests.

Frame 4 — Count the four adjacencies at THIS point

Focal color → Neighbor color

E neighbor: mid → mid (same)

W neighbor: mid → dark (transition)

N neighbor: mid → mid (same)

S neighbor: mid → mid (same)

Result: 1 transition (mid → dark) and 3 same-color pairs.

Frame 5 — These tallies fill in a transition matrix T

Transition matrix T — empty

focal color	dark	mid	light
dark	0	0	0
mid	0	0	0
light	0	0	0
neighbor color			

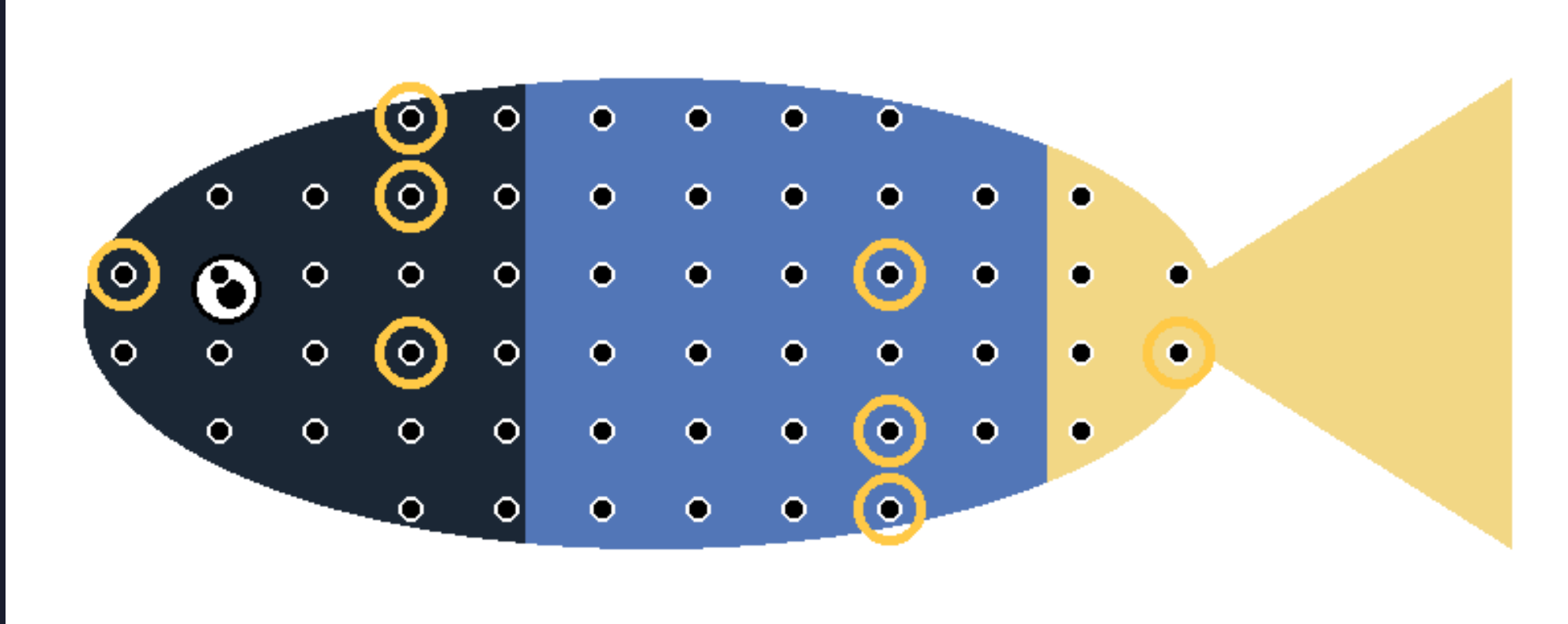
- Rows = focal color class.
- Columns = neighbor color class.
- Cell $T[i, j]$ = number of times focal color i was adjacent to neighbor color j .
- Diagonal cells ($T[i, i]$) = same-color adjacencies (= big blobs).
- Off-diagonal cells = color transitions (= edges).

Frame 6 — Increment cell $T[\text{mid}, \text{dark}]$ for the one transition



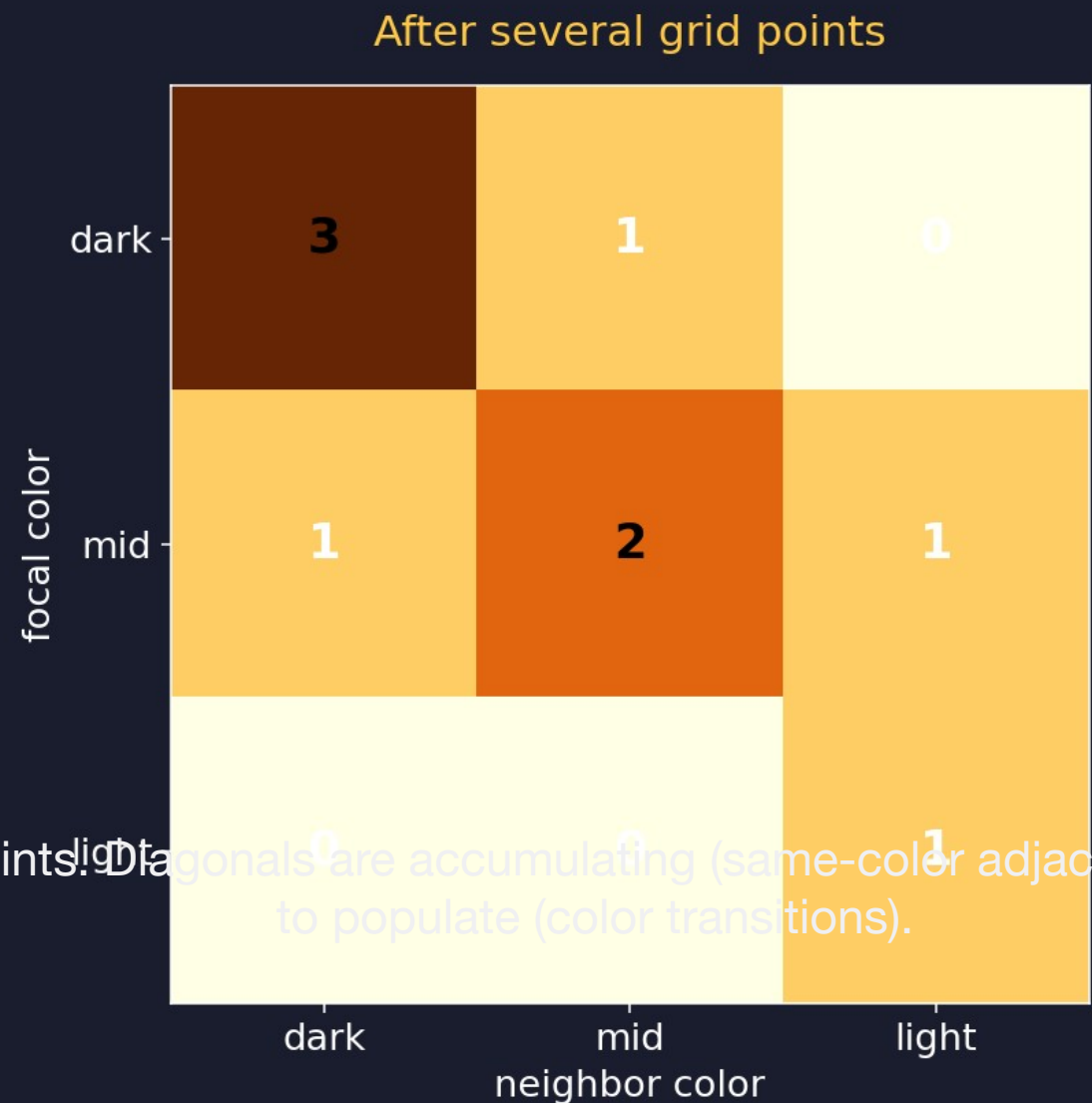
We're not actually incrementing $T[\text{mid}, \text{dark}] = 1$ yet — but conceptually, every adjacency we counted at this grid point contributes to ONE cell of T .

Frame 7 — Now do this at every grid point



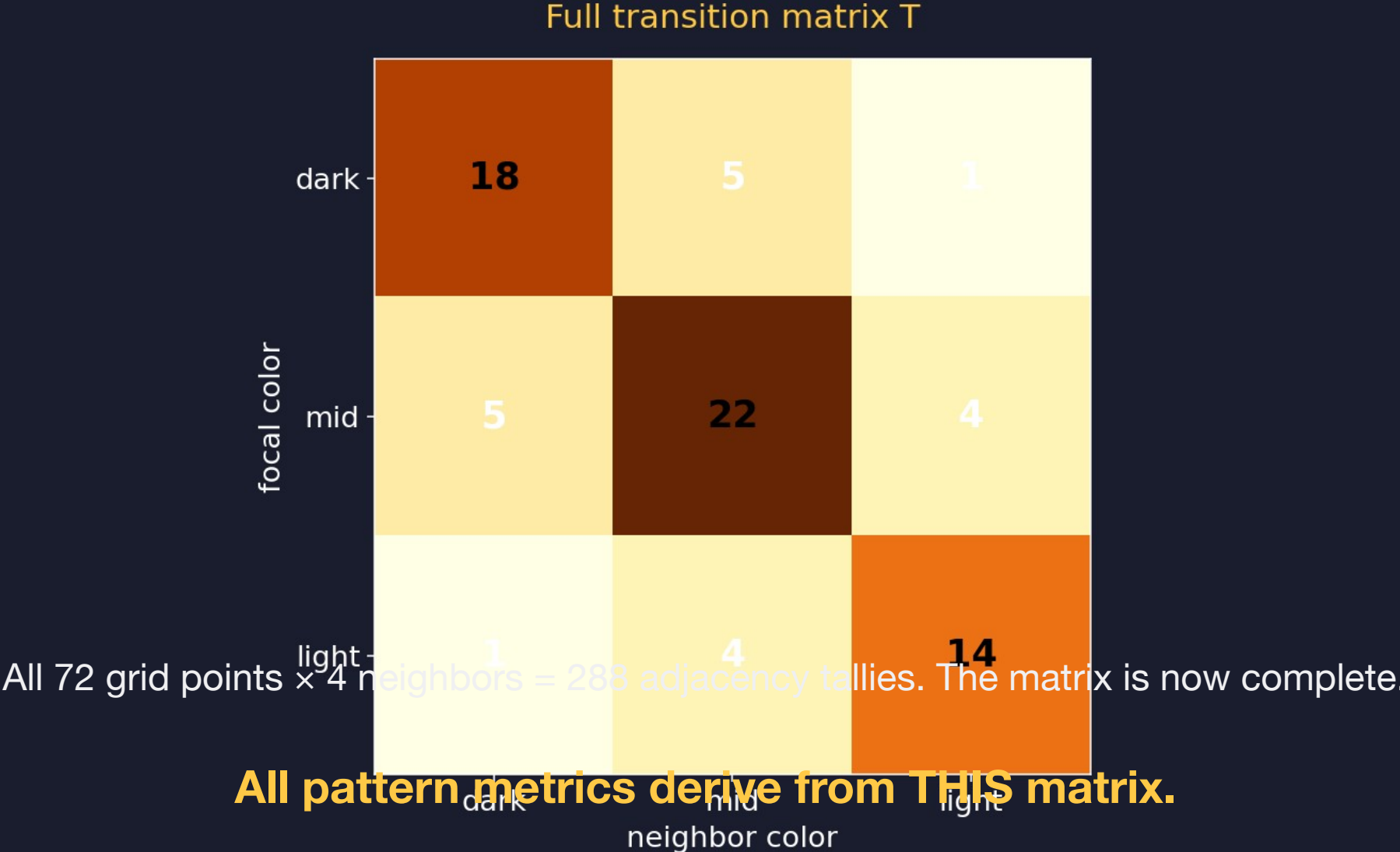
8 example focal points marked in gold. Same procedure at each: focal vs 4 neighbors, contribute 4 tallies to T.

Frame 8 — After several grid points, T fills in



Partial T after ~10 grid points! Diagonals are accumulating (same-color adjacencies). Off-diagonals starting to populate (color transitions).

Frame 9 — Full transition matrix T after all grid points



All of that — in one pavo function call

```
fish_adj <- adjacent(fish_k3,  
                     xpts  = 200,  
                     xscale = 5)
```

Output: a data frame of pattern statistics, computed from T.

xpts = number of grid points along the longest axis

xscale = pixel-to-real-world scale (cm) — used for distance metrics

Step 4 — The metrics

Six numbers that describe a pattern, each derived from T

The six metrics we'll focus on today

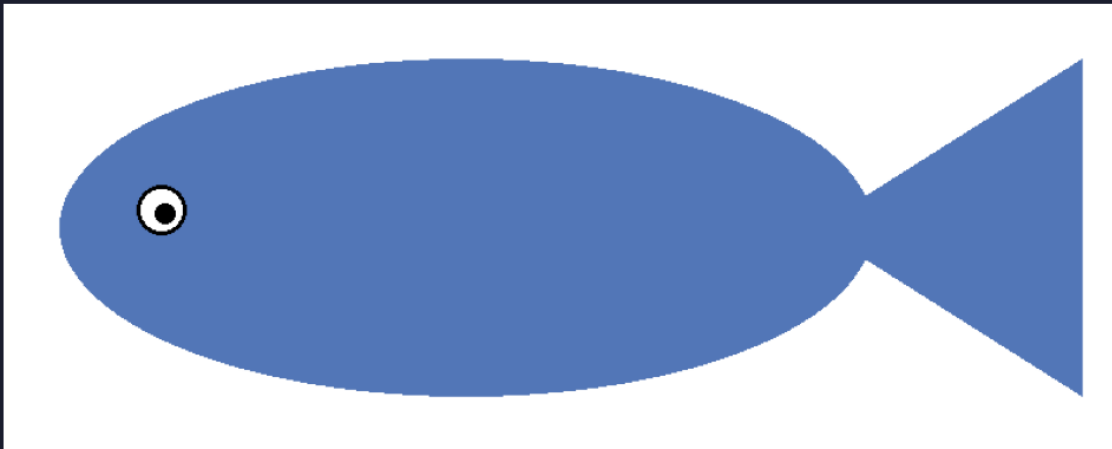
- m — transition density (off-diagonal mass / total)
- A — aspect ratio (vertical transitions vs horizontal)
- J_c — color heterogeneity (Simpson diversity of colors)
- J_t — transition heterogeneity (diversity of transition types)
- m_{dS} — chromatic boundary strength (saturation step at edges)
- m_{dL} — achromatic boundary strength (luminance step at edges)

Each one is a single number computed from T .

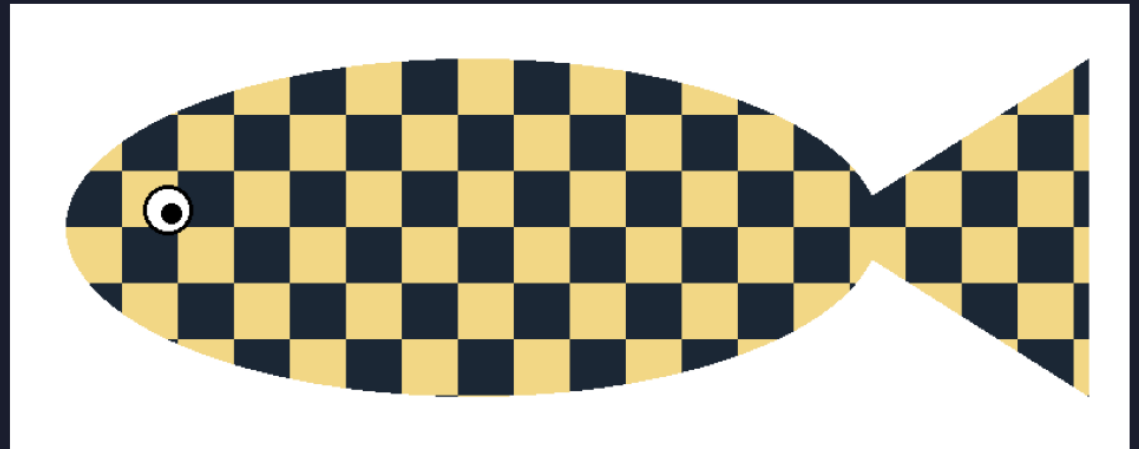
m — transition density

m — transition density (fraction of adjacent-pixel pairs that differ)

LOW m
Solid: every neighbor is the same color
($m \approx 0.00$)



HIGH m
Checkerboard: every neighbor is different
($m \approx 0.95$)



Low m = mostly same-color neighbors → big blobs. High m = lots of color changes → busy pattern.

A — aspect ratio

A — aspect ratio (orientation of edges; deviation from 1 = directional)

LOW A

Horizontal bands (toy): transitions run mostly L↔R
($A \approx 0.4$)



HIGH A

Sergeant major: vertical bars → vertical transitions dominate
($A \approx 2$)



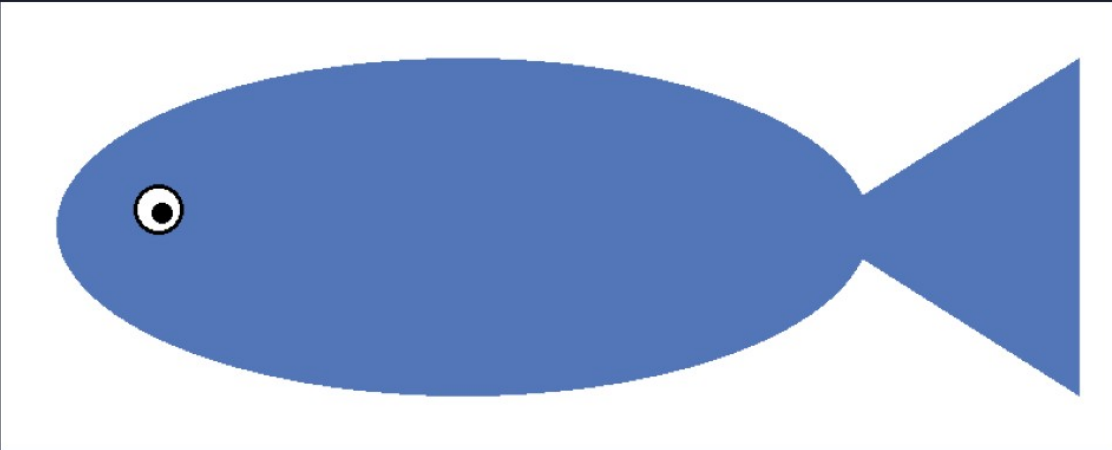
Pavo's $A = (\text{vertical transitions}) / (\text{horizontal transitions})$. Toy horizontal bands → $A < 1$ (low). The real sergeant major has vertical bars → $A > 1$ (high). Reticulate / spotted patterns sit near $A \approx 1$.

Jc — color heterogeneity

Jc — color heterogeneity (effective number of color classes)

LOW Jc

Solid (toy): only 1 color; one diagonal entry dominates
($Jc \approx 0.00$)



HIGH Jc

Mandarin fish: many colors, evenly mixed
($Jc \approx 0.7$)

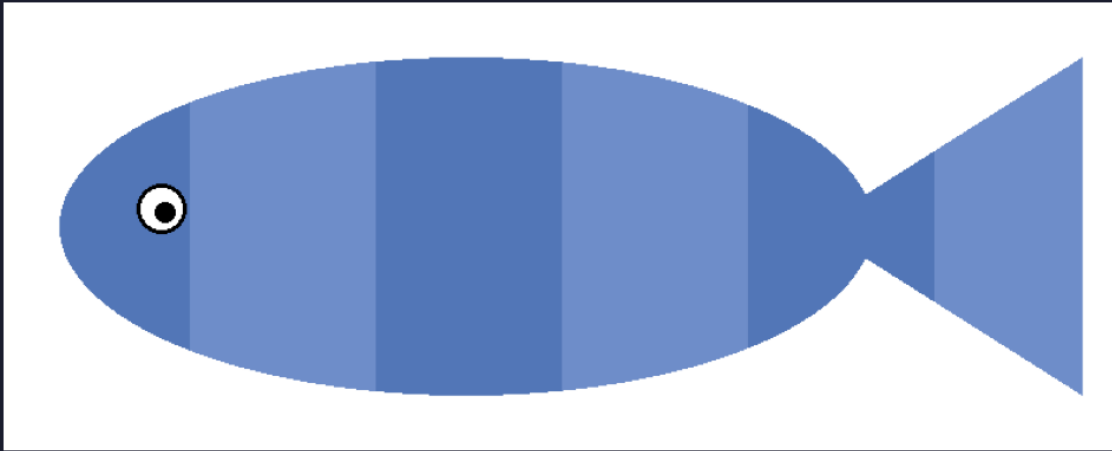


Jc tracks how many color classes show up and how evenly they're used. Toy solid fish → low Jc. The real mandarin fish has many colors mixed evenly → high Jc.

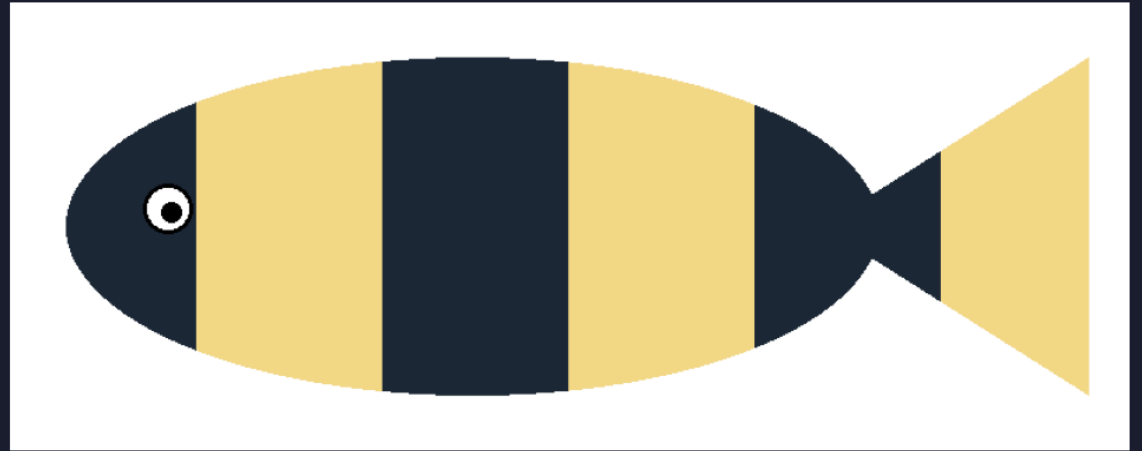
m_dL — luminance boundary

m_dL — achromatic boundary strength (mean luminance step at edges)

LOW m_dL
Adjacent colors are similar → small ΔL
(m_dL $\approx$ small)



HIGH m_dL
Adjacent colors are dark/bright → large ΔL
(m_dL $\approx$ large)



How bright is the contrast across edges? Black-on-white = strong achromatic boundary. Two similar greens = weak.

Predict, then measure

Before running pavo on the three demo fish —
what would you expect each metric to look like?

Predict — one minute, with your neighbor

Predict before you measure

Fish	Family	Pattern type	Predicted	Predicted A
Sergeant major	Pomacentridae	vertical bars	high m, $A > 1$	≈ 2
Cleaner wrasse	Labridae	horizontal stripe	low m, $A < 1$	≈ 0.5
Mandarin fish	Callionymidae	reticulate	highest m, $A \approx 1$	≈ 1

Take 60 seconds. Discuss with your neighbor: do you agree with these predictions? Where might they be wrong?

Now run the live demo

Switch to RStudio.

`scripts/walkthrough.R`

We'll process all three fish, build a comparison table, and check our predictions.

Now — your turn

Pick THREE fish from your assigned lineage — three DIFFERENT pattern types.
Run all three. Compare to the demo fish.

Your assignment for today

- Your group has an assigned reef fish lineage (typically a family).
- Pick THREE species from your lineage that look as different from each other as you can find:
 - one with vertical bars (you predict $A > 1$)
 - one with horizontal stripes or bands (you predict $A < 1$)
 - one with spotty / reticulate / solid pattern (you predict $A \approx 1$)
- Run the full pipeline on each: getimg → classify → adjacent → summary.
- Compare your three A values to the demo fish you just saw.

Did pavo's A statistic classify your fish the same way it classified the demos?

How to pick your three fish

- Use whatever image source your group has been working with — FishView, FishBase, your own field photos, or the fallback folders in `images/student-fish/<your-lineage>/`.
- Look at the photos before you classify. Pick three that span the widest pattern range your lineage offers.
- If your lineage doesn't have all three pattern types (e.g., all surgeonfishes are vaguely solid-colored), don't force it — just pick the three with the most different patterns you can find.
- REGISTER each image to LEFT-LATERAL view (head on left). If a fish is facing the wrong way, flip it horizontally with `flip_to_left_lateral()` or `magick -flop`.
- Talk to your group; aim for different choices across the four members so you have more data to compare in the wrap-up.

What you'll fill in

Three metric tables, plus a comparison to the demo fish.

Example: your worksheet's results table

Metric (reticulate)	Fish 1 (bars)	Fish 2 (stripe)	Fish 3
k	3	3	5
m	0.13	0.07	0.28
A	2.4	0.45	1.05
Jc	0.62	0.48	0.78
m_dL	0.55	0.30	0.40

Then ONE comparison question (do your fish track the demo pattern?)
and ONE biological interpretation (selection hypothesis for one fish).

Wrap-up — 5 min

Compare results across groups. Predict for next session.

What we did today

- Loaded reef fish photos with `getimg()`.
- Reduced millions of RGB values to k color classes with `classify()`.
- Sampled the pattern with a regular grid (Endler 2012).
- Counted color transitions in a transition matrix T .
- Derived 6 pattern metrics from T with `adjacent()`.
- Predicted, measured, and compared three demo fish.
- Each of you ran your own assigned fish through the same pipeline.

Next time

- We will run pavo on multiple species per family.
- Compute a metric MATRIX — species $\times$ metric.
- Cluster species by pattern. Compare to phylogeny.
- Ask: do related fish share patterns? Or has selection pushed lineages to different parts of pattern space?

Today was the toolkit. Next time we use it to do biology.

Questions?

Submit your worksheet to BruinLearn by end of day.

Files: images/ • scripts/walkthrough.R • lab-05-pavo-intro.qmd